



DSA Day 3 Preparation

Learning Data Structures and Algorithms (DSA) using Python and Java



Day 3 focuses on loops, repetition, and loop-based problem solving in Python and Java.



1 Topics Covered

- What is a Loop?
- Why Loops are Important?
- For Loop
- While Loop
- For Loop vs While Loop
- Infinite Loop
- Break Statement
- Continue Statement
- Time Complexity of Loops
- Loop-Based Practice Problems
- LeetCode Practice Problems

2 What is a Loop?

A loop is used to repeat the same task again and again.

Example: Print numbers from 1 to 5

1 2 3 4 5



Without loops, we would have to write five print statements. Loops help us do this with fewer lines of code.

3 Why Loops are Important?



- Find maximum number
- Count even numbers
- Search for a value
- Calculate sum
- Print all elements

In DSA, loops help us process multiple values one by one.

4 For Loop

Python

```
for i in range(1, 6):  
    print(i)
```



Java

```
for (int i = 1; i <= 5; i++) {  
    System.out.println(i);  
}
```



For loop is used when repetitions are known.

5 While Loop

Python

```
i = 1  
while i <= 5:  
    print(i)  
    i += 1
```



Java

```
int i = 1;  
while (i <= 5) {  
    System.out.println(i);  
    i++;  
}
```



Use while loop when repetition depends on a condition.

6 For Loop vs While Loop

	For Loop	While Loop
	repetitions known	repetition depends on condition
	common for counting	common for condition-based repetition
	example: print 1 to n	example: repeat until input is valid

7 Infinite Loop



Happens when the stopping condition never becomes false.



Example:
while (True):
 pass

8 Break Statement

```
for i in range(1, 6):  
    if i == 3:  
        break  
    print(i)
```

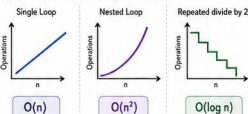
Stops a loop immediately.

9 Continue Statement

```
for i in range(1, 6):  
    if i == 3:  
        continue  
    print(i)
```

Skips the current iteration.

10 Time Complexity of Loops



11 Practical Problems

- ✓ Print Numbers from 1 to N
- ✓ Print Even Numbers from 1 to N
- ✓ Sum of First N Natural Numbers
- ✓ Factorial of a Number
- ✓ Count Digits in a Number

12 LeetCode Practice Problems for Day 3

#	Problem Title
1	1342. Number of Steps to Reduce a Number to Zero
2	412.Fizz Buzz
3	258. Add Digits
4	1295. Find Numbers with Even Number of Digits
5	1480. Running Sum of 1d Array

★	Best Problems to Solve First
1	1342. Number of Steps to Reduce a Number to Zero
2	412. Fizz Buzz
3	1295. Find Numbers with Even Number of Digits

13 Day 3 Summary



- ✓ loops help repeat code
- ✓ for loop when repetitions are known
- ✓ while loop when repetition depends on condition
- ✓ break stops a loop
- ✓ continue skips one iteration
- ✓ single loop usually $O(n)$
- ✓ nested loop usually $O(n^2)$
- ✓ repeated division can be $O(\log n)$

14 Learning Journey



Day 1
Programming
Basics,
Complexity,
Constraints



Day 2
Conditions
and Decision
Making



Day 3
Loops and
Loop-Based
Problems



Day 4
Pattern
Problems and
Nested Loops

→ This repository contains my Day 3 notes for learning DSA using python and Java.

→ The main goal of Day 3 is to Understand loops, because loops are used to repeat work and process multiple values.

① What is a loop?

→ A loop is used to repeat the same task again and again.

Example:

Print numbers from 1 to 5

without loop:

1, 2, 3, 4, 5

→ if we want to print numbers from 1 to 100, writing 100 print statements is not practical.

→ So we use a loop. A loop help us repeat code easily.

② Why loops are important?

→ Loops are very important in programming and DSA.

→ In DSA, we often work with many values.

Example:

→ Find maximum number

→ Count even numbers.

→ Search for a value.

→ Calculate sum.

→ Print all elements

→ To do these tasks, we need to visit values one by one.
That is why loops are important.

③ For loop.

→ A for loop is used when we know how many times we want to repeat something.

Example:

→ print numbers from 1 to 5.

Python:

```
for i in range(1, 6):  
    print(i)
```

Output:

1

2

3

4

5

Java:

```
class Main {
```

```
    public static void main(String[] args) {
```

```
        for (int i = 1; i <= 5; i++) {
```

```
            System.out.println(i);
```

```
        }
```

```
    }
```

→ range(1, 6) gives numbers from 1 to 5.

→ 6 is not included.

→ i takes one value at a time.

→ int i = 1 → starting value

→ i <= 5 → condition

→ i++ → increase by 1

④ while loop:

A while loop runs as long as the condition is true
Use while loop when the number

Python:

```
i = 1
while i <= 5:
    print(i)
    i += 1
```

output:

1
2
3
4
5

Java

```
class Main {
    public static void main (String[] args) {
        int i = 1;
        while (i <= 5) {
            System.out.println (i);
            i++;
        }
    }
}
```

important:

- Always update the loop variable.
- Otherwise, the loop may never stop.

⑤ for loop (vs) while loop.

for loop

- Used when repetitions are known.
- Common for counting.
- Example: print 1 to n.

while loop

- used when repetition depends on condition.
- Common for condition-based repetition.
- Example: repeat until input is valid.

Simple rule:

- Use for loop when you know the range.
- Use while loop when you only know the stopping condition.

⑥ Infinite Loop:

→ An infinite loop is a loop that never stops.

Example:

```
i = 1
```

```
while i <= 5:
```

```
    print(i)
```

→ Here, i is never increased. → output (infinite 1's)

so $i <= 5$ is always true.

Correct version

```
i = 1
```

```
while i <= 5:
```

```
    print(i)
```

```
    i += 1
```

⑦ Break Statement:

→ break is used to stop a loop immediately.

→ Python

```
for i in range(1, 6):
```

```
    if i == 3:
```

```
        break
```

```
    print(i)
```

Output:

1
2

Java:

```
class Main {
```

```
    public static void main (String[] args) {
```

```
        for (int i = 1; i <= 5; i++) {
```

```
            if (i == 3) {
```

```
                break;
```

```
            }
```

```
            System.out.println (i);
```

```
        }
```

```
    }
```

```
}
```

⑧ Continue statement:

→ Continue skips the current iteration and moves to the next one.

Python:

```
for i in range (1,6):
```

```
    if i == 3:
```

```
        continue
```

```
    print(i)
```

Output:

1

2

4

5

Java:

```
class Main {
```

```
    public static void main (String[] args) {
```

```
        for (int i = 1; i <= 5; i++) {
```

```
            if (i == 3) continue;
```

```
            System.out.println (i);
```

Q. Time Complexity of Loops

→ Loops directly affect time complexity

Single Loop:

```
for i in range(n):  
    print(i)
```

→ if $n=5$, loop runs 5 times.

→ if $n=100$, loop runs 100 times.

So time complexity is: $O(n)$

Nested Loop:

```
for i in range(n):  
    for j in range(n):  
        print(i, j)
```

Outer loop runs n times.

Inner loop runs also n times.

⇒ $n \times n = n^2 \rightarrow$ Time complexity: $O(n^2)$

Loops that divide by 2.

```
while n > 1:  
    n = n // 2
```

Here, n is divided by 2 each time.

Time complexity:

$O(\log n)$